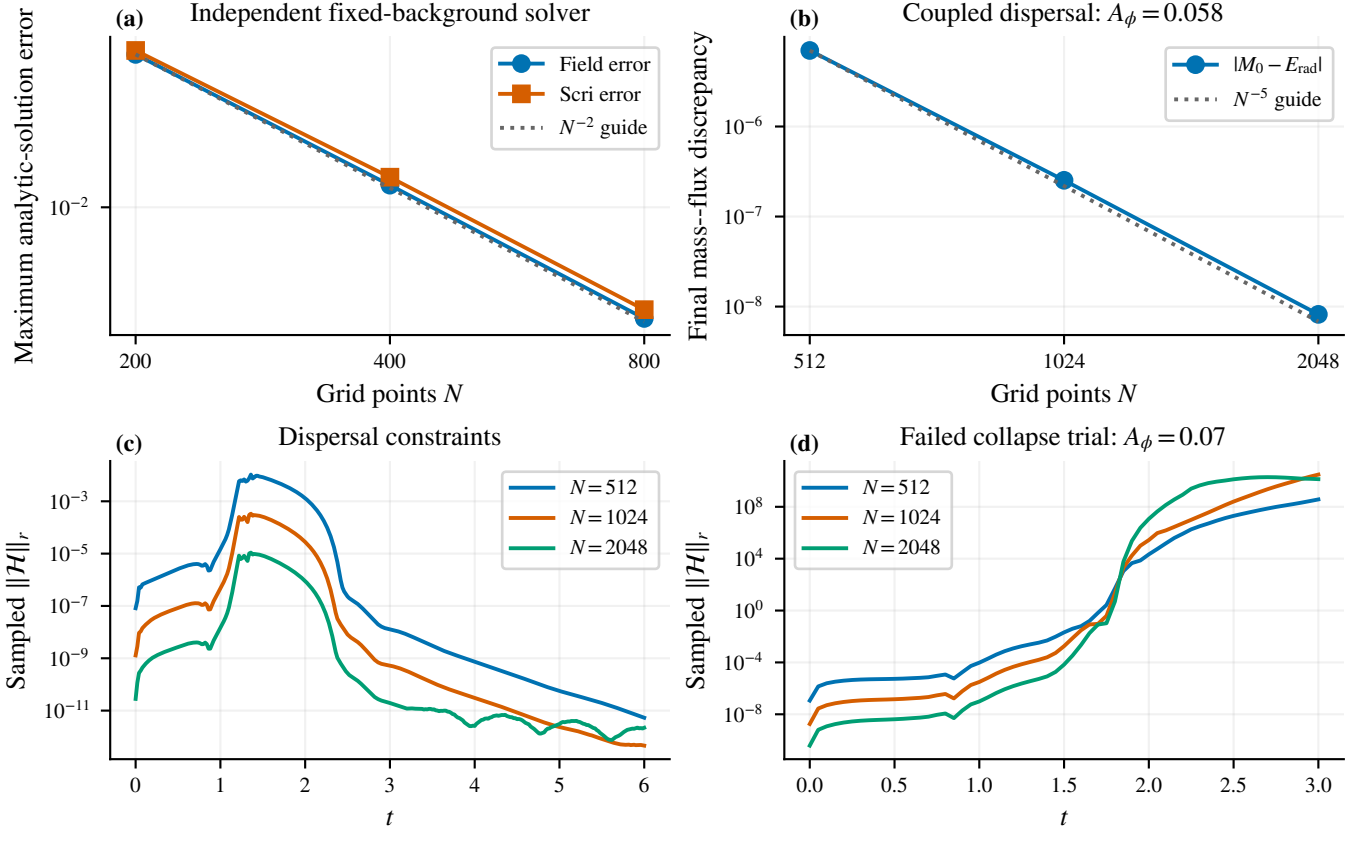
(a)
Independent fixed-background solver
Maximum analytic-solution error
Field error
Scri error
N^{-2} guide
10^{-2}
200
400
800
Grid points N
(b)
Coupled dispersal: $A_\phi = 0.058$
Final mass-flux discrepancy
$|M_0 - E_{\rm rad}|$
N^{-5} guide
10^{-6}
10^{-7}
10^{-8}
512
1024
2048
Grid points N
(c)
Dispersal constraints
Sampled $\|\mathcal{H}\|_r$
$N = 512$
$N = 1024$
$N = 2048$
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
0
1
2
3
4
5
6
t
(d)
Failed collapse trial: $A_\phi = 0.07$
Sampled $\|\mathcal{H}\|_r$
$N = 512$
$N = 1024$
$N = 2048$
10^{8}
10^{4}
10^{0}
10^{-4}
10^{-8}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
t